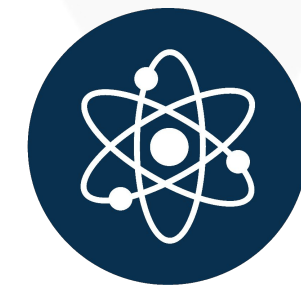


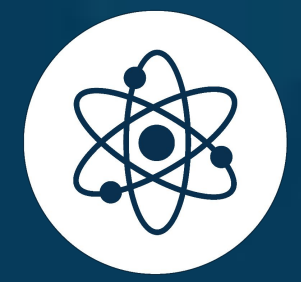
EXPERIMENT TO VERIFY BOYLE'S LAW

PROFESSOR AKINOLA OLAYINKA,
PhD Physics



AIM

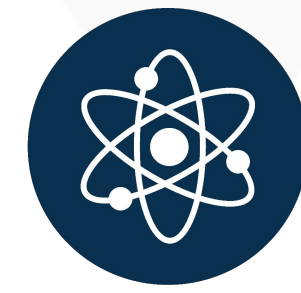
To verify Boyle's law.



LEARNING OBJECTIVES

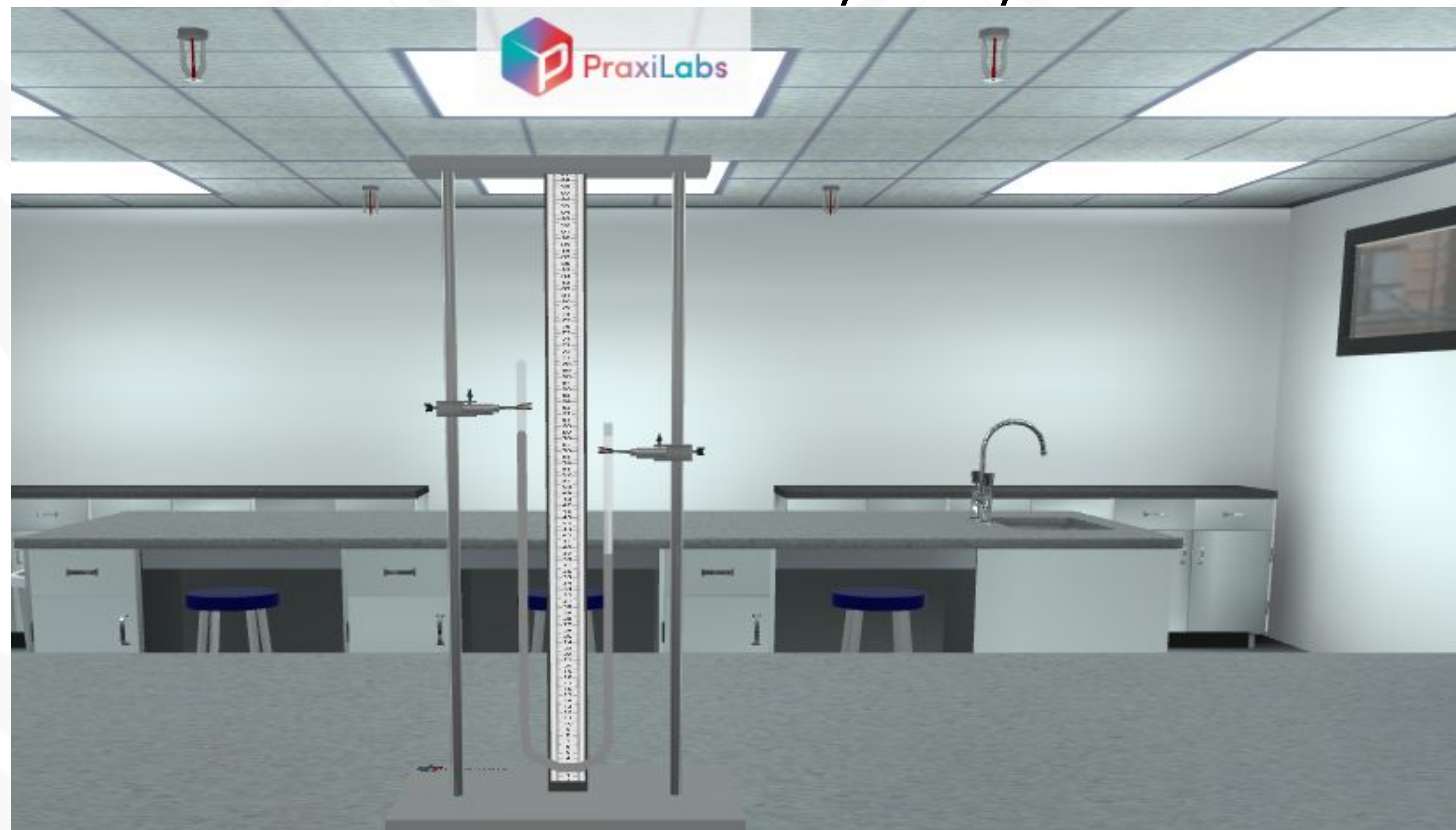
At the end of this lesson, you should be able to:

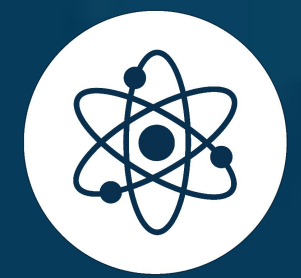
- Define pressure.
- Identify units for measuring pressure.
- Convert pressure and volume values between units.
- Explain the pressure-volume relationship in ideal gases.
- Design an experiment to investigate the pressure-volume relationship in ideal gases.



APPARATUS

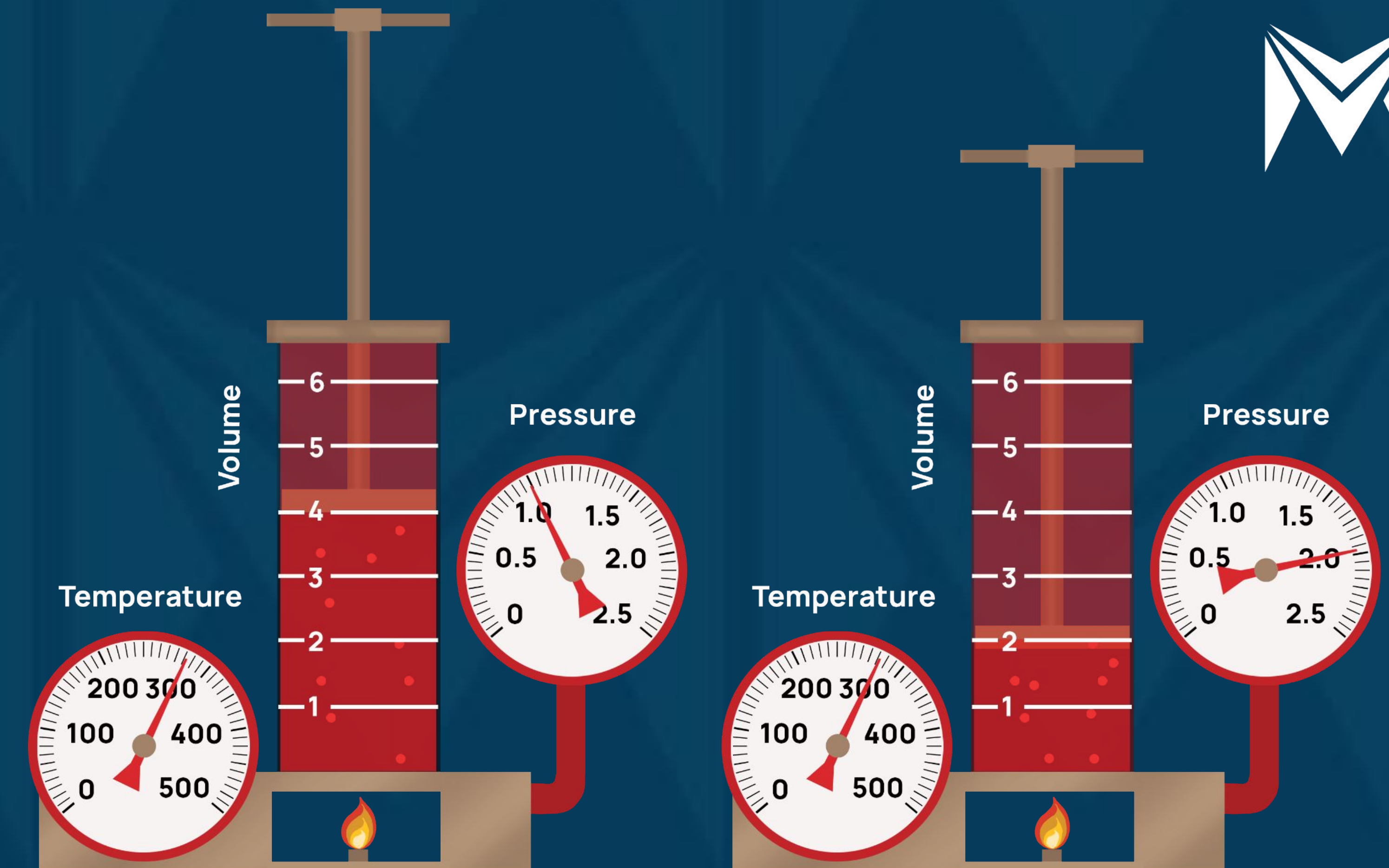
- J-tube
- Thermometer
- Pressure Sensor
- Gas Sample
- Praxilabs virtual laboratory – Boyle's Law

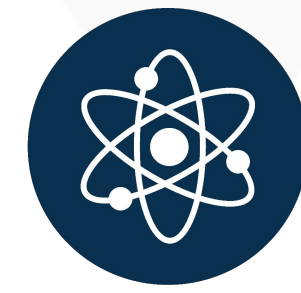




THEORY

- Boyle's law, along with other empirical laws like Charles's law, Avogadro's law, and Gay-Lussac's law, played a pivotal role in shaping the ideal gas law, which forms the foundation of the kinetic theory of gases.
- Although the law was independently discovered by French physicist Edme Mariotte in 1679, it is commonly attributed to Robert Boyle, who published his findings in 1662. This law is sometimes referred to as the Boyle-Mariotte law.

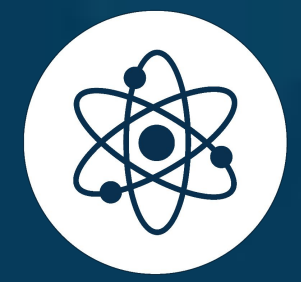




THEORY

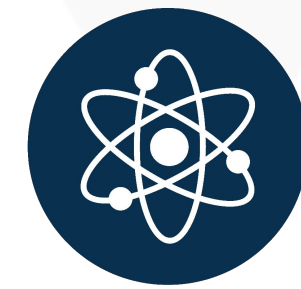
- Boyle's law describes the inverse relationship between the pressure and volume of a fixed quantity of gas at a constant temperature.
- It states that the product of the pressure (P) and volume (V) of a gas is constant as long as the temperature remains constant.
- Or At constant temperature, the pressure of a definite quantity of an ideal gas is inversely proportional to its volume.
- Mathematically, Boyle's law can be expressed as:

$$PV = \text{Constant}$$
$$\text{or } P \propto 1/V$$



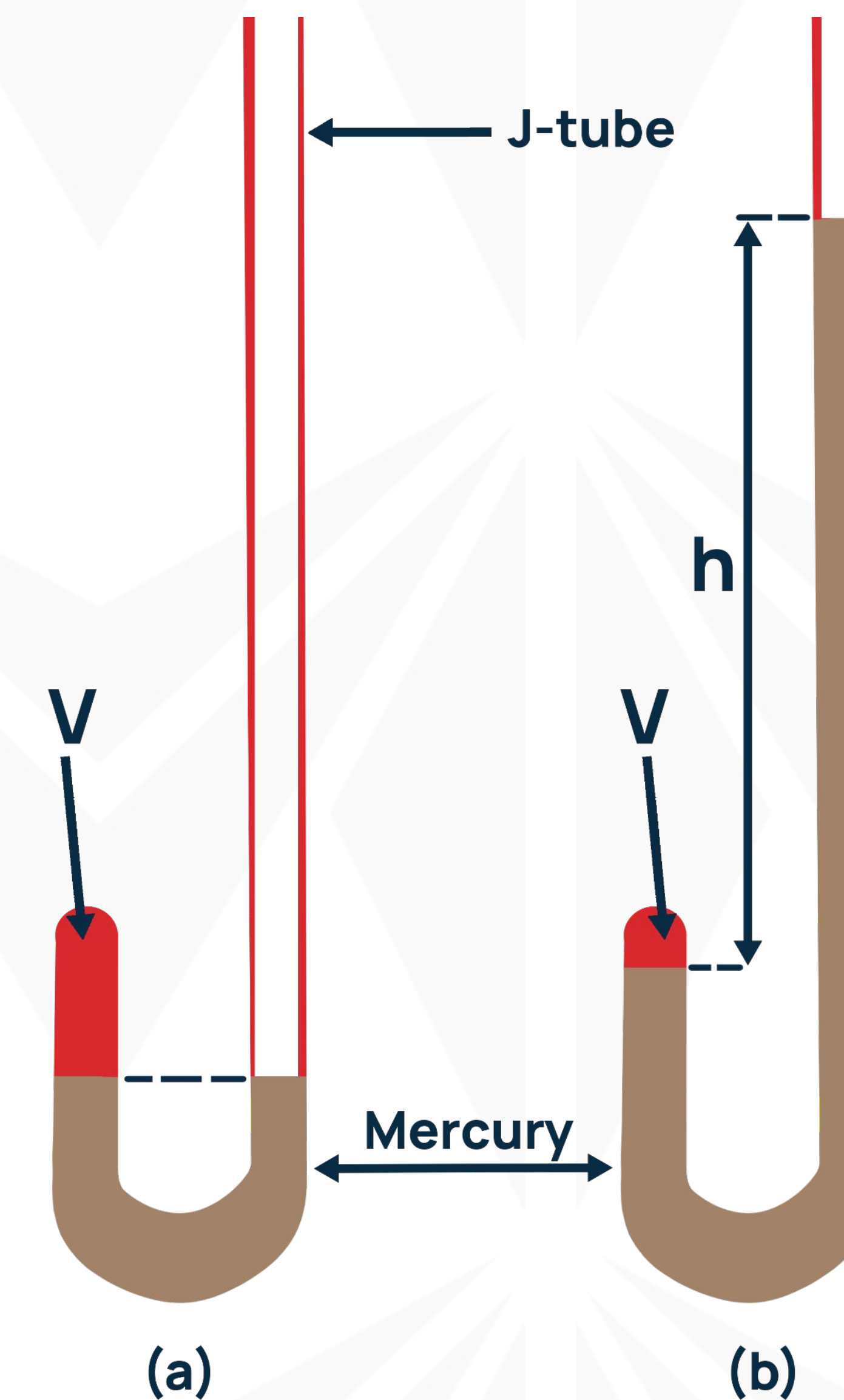
THEORY

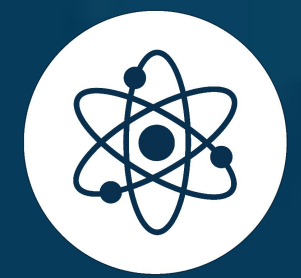
- The pressure is defined as the force F acting per unit area A . $P = F/A$
- The SI unit of pressure is Pascal ($\text{Pa} = \text{N/m}^2$).
- Recall that, at sea level, atmospheric pressure supports a column of mercury 76 cmHg (or 760 mmHg) high, and it is related to the Pascal by:
 $1 \text{ atm} = 76 \text{ cm Hg} = 1.013 \times 10^5 \text{ Pa}$



THEORY

- Now consider the J-tube as shown in Figure (a), which has one closed end, and the other end is open to the air.
- If mercury is poured into this tube from its open end, a certain quantity of air is trapped in the closed tube.
- If the pressure exerted on the trapped air resulting from the mercury in the closed tube is equal to the atmospheric pressure, mercury levels in both tubes will be the same.

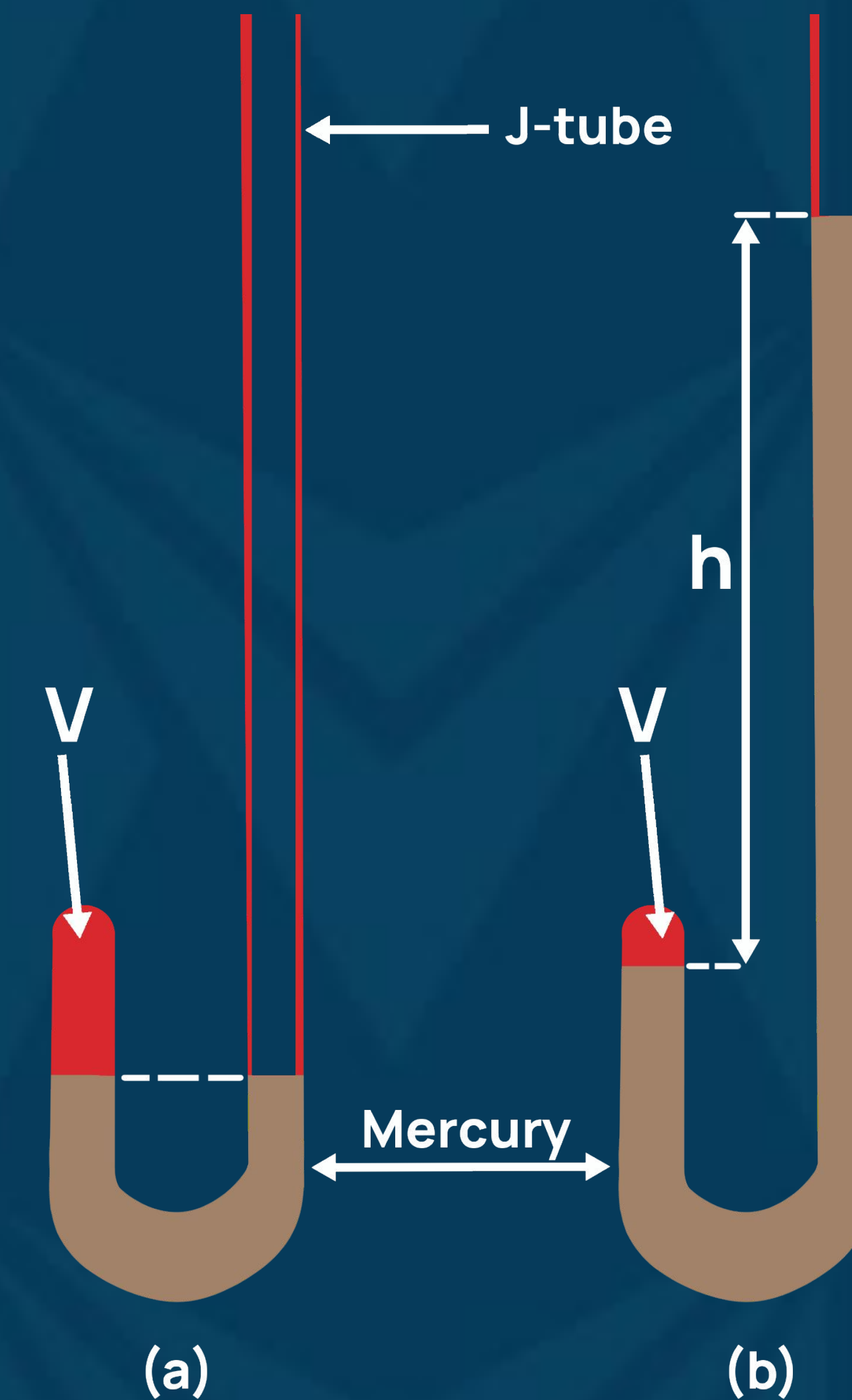


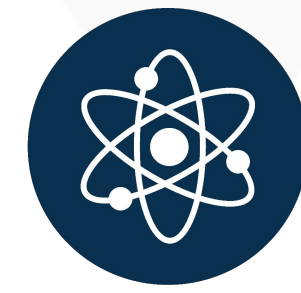


THEORY

- Now if we have raised the open-end tube or poured more mercury into it, the pressure on the trapped air increases by the amount Δp made by the excess mercury in the open tube.
- That is, the pressure is given by: $p_f = p_a + \rho g h$
where $\rho = 13600 \text{ kg/m}^3$ is the density of mercury and $g = 9.8 \text{ m/s}^2$ is the free-fall acceleration.

$$p_f = p_a + \rho g h \rightarrow \Delta p = p_f - p_a = \rho g h$$





THEORY

- If initially a certain amount of air is trapped inside the closed-end tube,

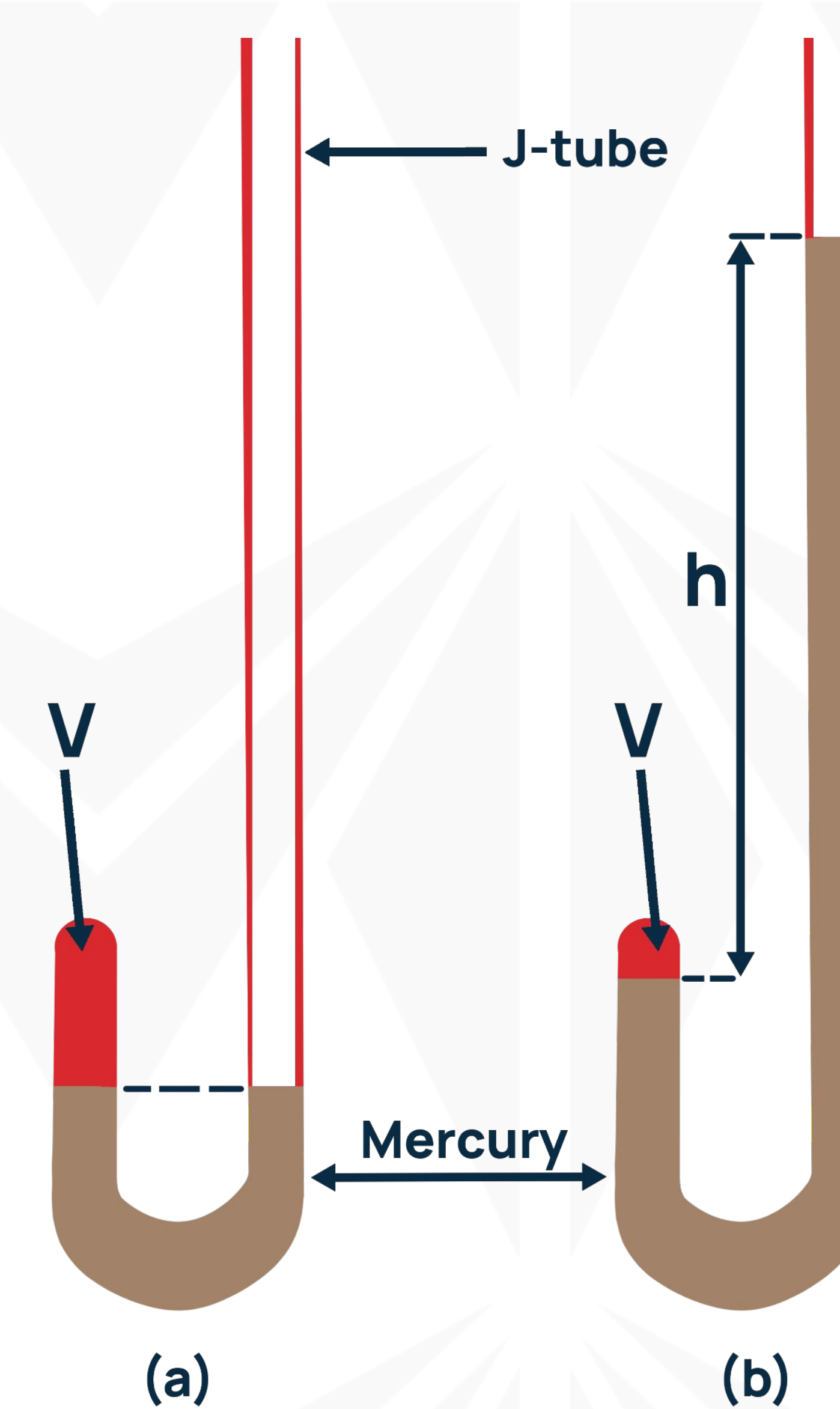
- say $V_o = 20 \text{ ml}$, while

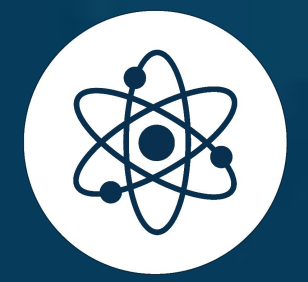
$$P_o = P_a = 76 \text{ cmHg} = 1.013 \times 10^5 \text{ Pa}$$

- Therefore,

$$P_o V_o = 1.013 \times 10^5 \text{ Pa} \times 20 \times 10^{-6} \text{ m}^3 \\ = 2.026 \text{ J}$$

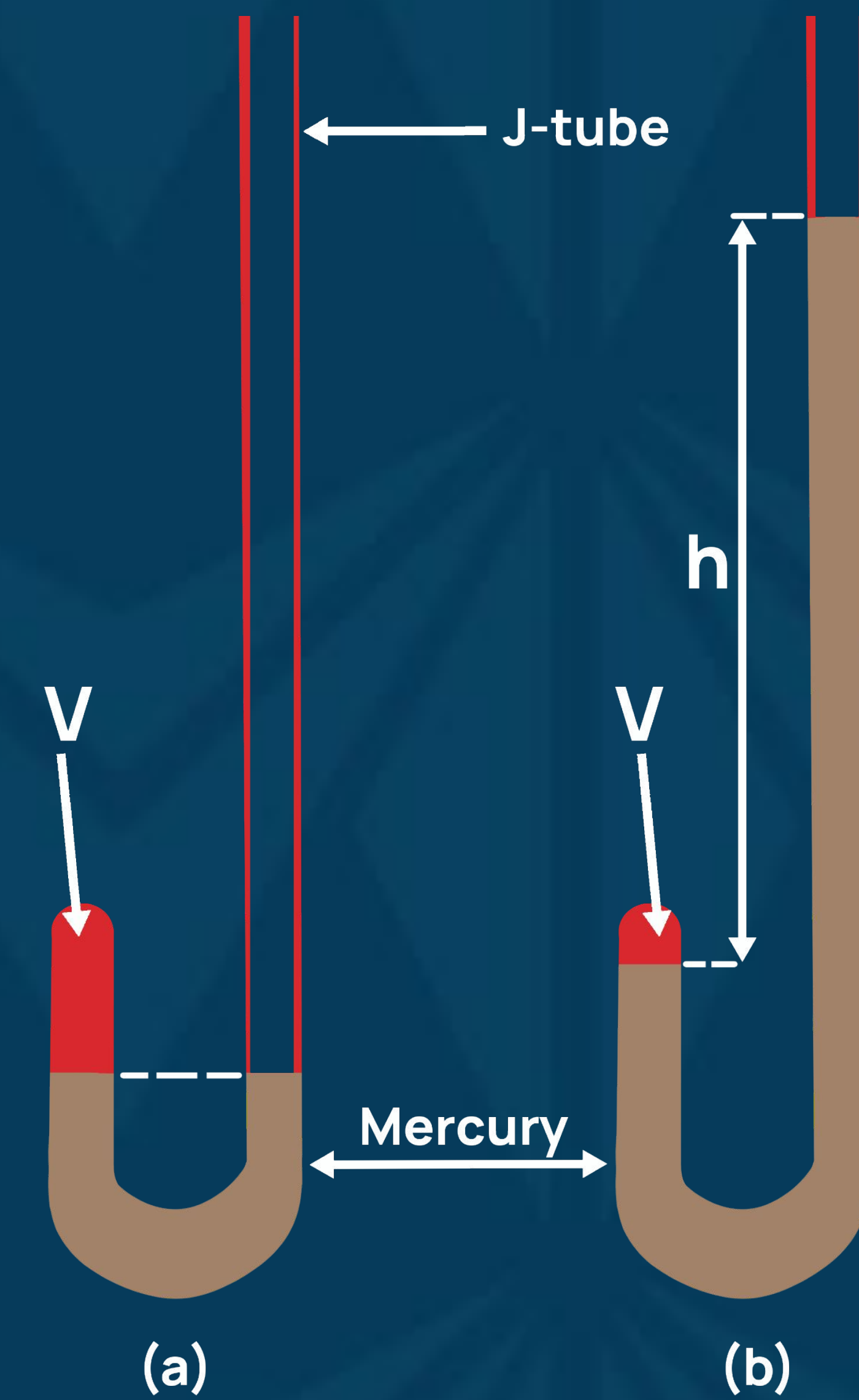
$$(1 \text{ milliliter (ml)} = 1 \times 10^{-6} \text{ cubic meters (m}^3))$$

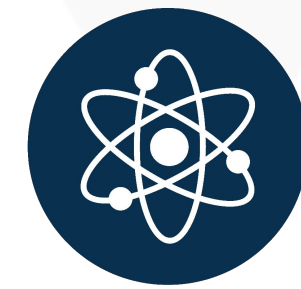




THEORY

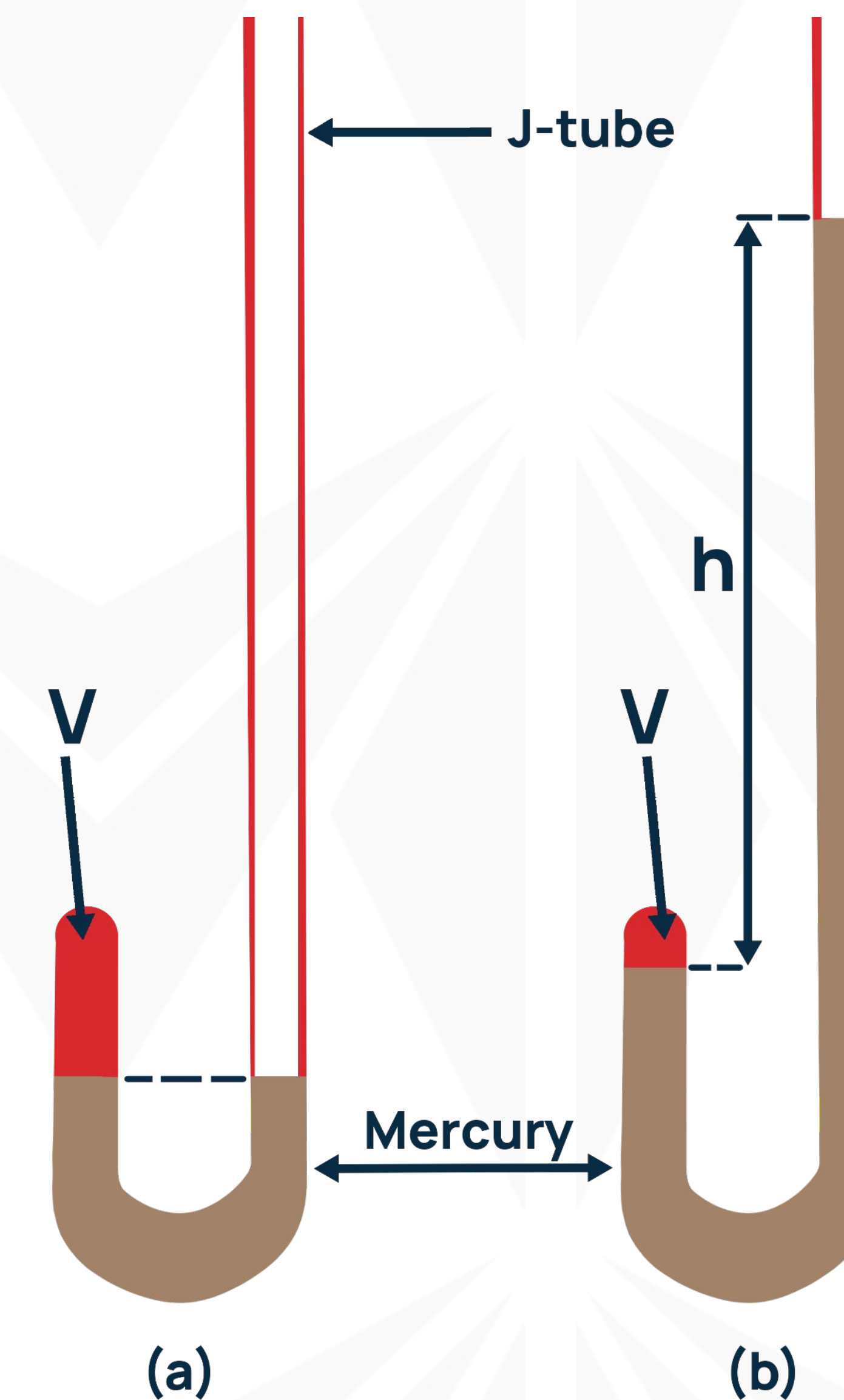
- Hence, the pressure corresponding to the volume occupied by the trapped air on moving the open-end tube a distance h upward can be calculated using
- $P_f = P_a + \rho gh$
- $= (1.013 \times 10^5) + (13600 \times 9.8 \times h)$

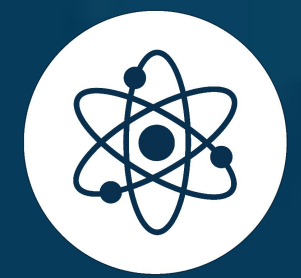




THEORY

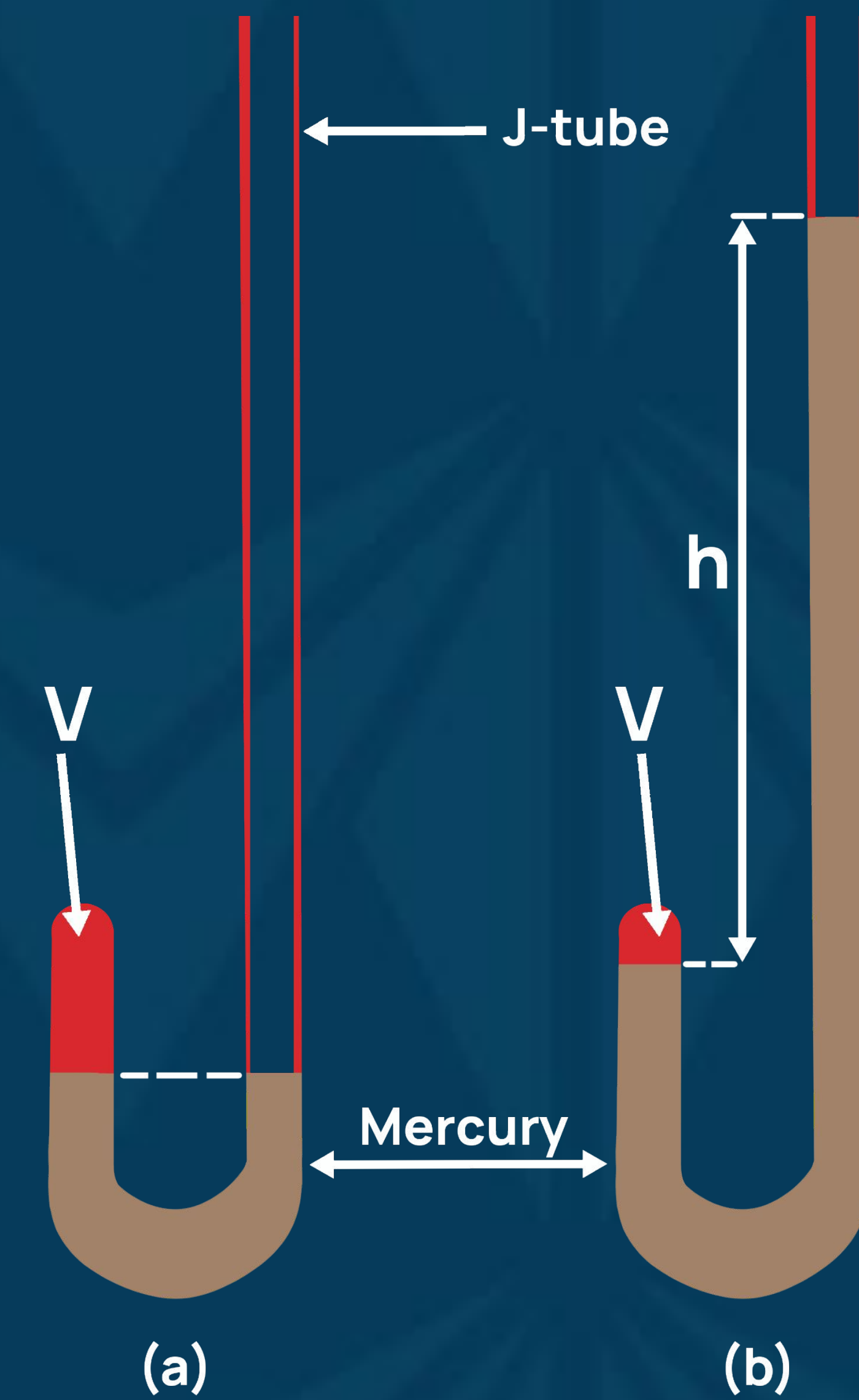
- On the other hand, the volume of the trapped gas can be found by measuring the length of the air column trapped inside the closed tube ℓ , which is identical with the volume of air, so
 - $V_f = \ell A$
where A is the area of the tube, which equals 1 cm^2 so, $V_f = \ell$
 - The new values for the pressure and volume are related to the original values by Boyle's law:
 - $p_f V_f = p_0 V_0 = 2.026$
 - Or $1/v_f = p_f/2.026$

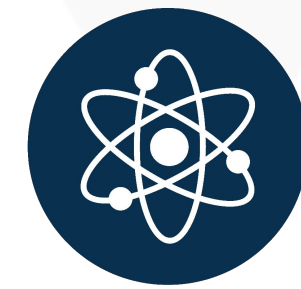




THEORY

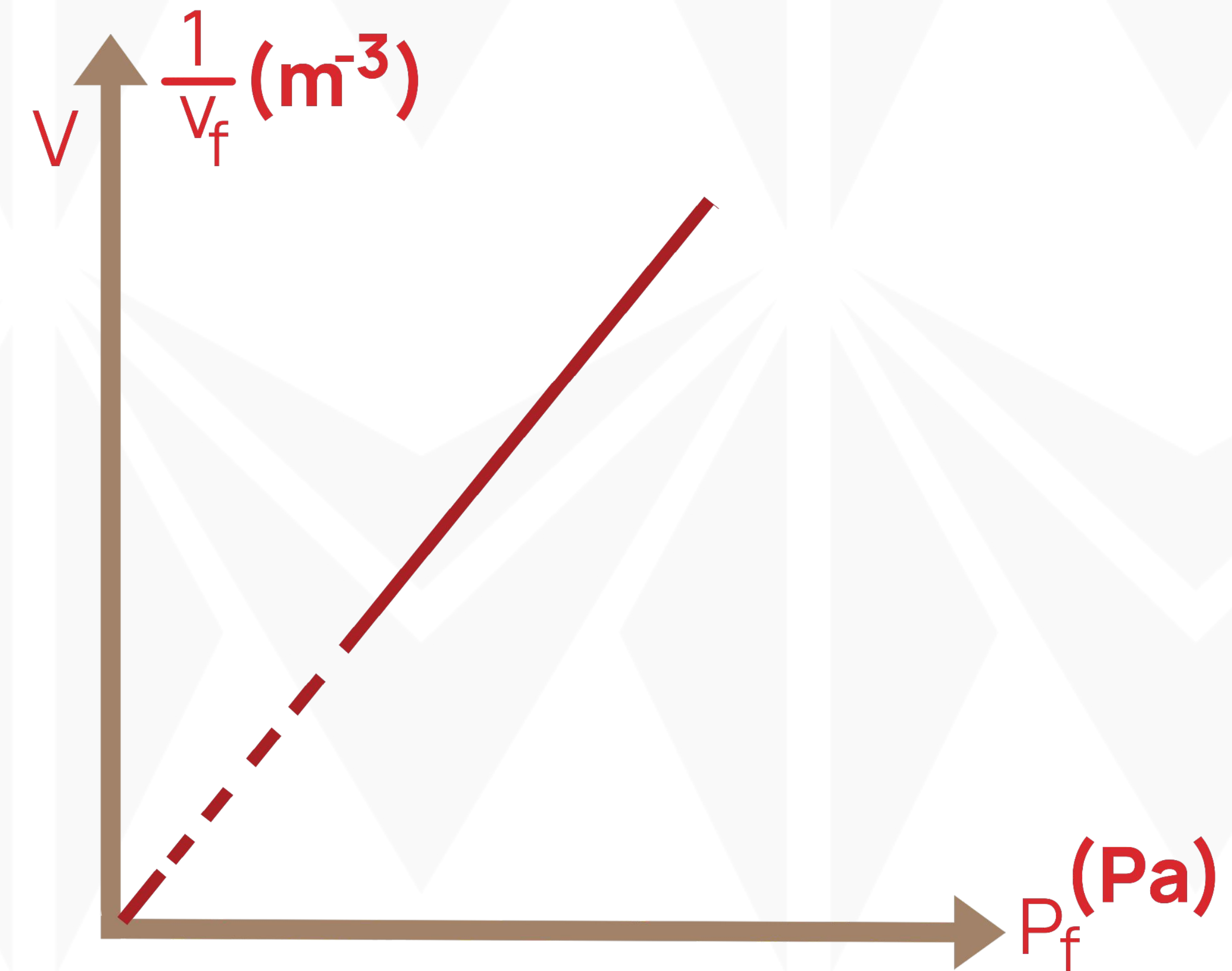
- ✓ The new values for the pressure and volume are related to the original values by Boyle's law:
- ✓ $p_f V_f = p_o V_o = 2.026$
- ✓ Or $\frac{1}{v_f} = \frac{p_f}{2.026}$
- ✓ Compare with Linear equation:
 $y = mx + c$

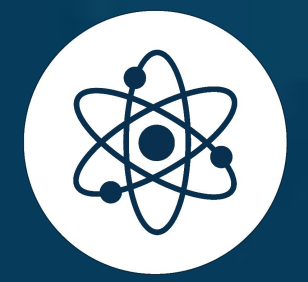




THEORY

- ✓ Now if we plot a graphical relation between $1/V_f$ and p_f
- ✓ we should get a straight line as shown.
- ✓ Hence, Boyle's law is verified.

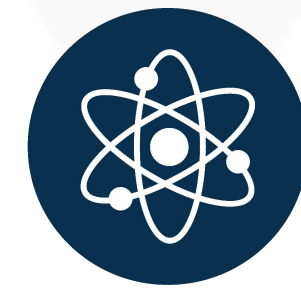




MID-LESSON QUESTIONS

Question 1: What does Boyle's Law state?

Question 2: How can Boyle's Law be expressed mathematically?



MID-LESSON ANSWERS

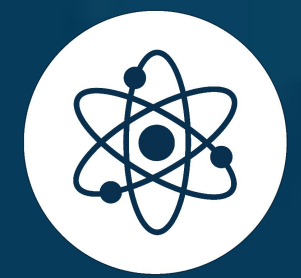
Answer 1: Boyle's Law states that the pressure and volume of a fixed quantity of gas at a constant temperature are inversely proportional. In other words, when the volume of a gas increases, its pressure decreases, and vice versa, as long as the temperature remains constant.

Answer 2: Boyle's Law can be expressed mathematically as:

$$P_1 \cdot V_1 = P_2 \cdot V_2$$

where:

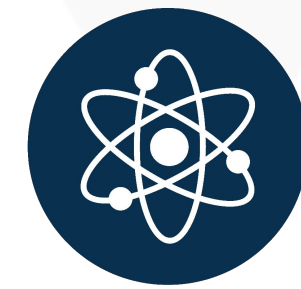
- P_1 and V_1 are the initial pressure and volume.
- P_2 and V_2 are the final pressure and volume.



PROCEDURE



1. Start with the mercury being at the same level on both sides. ($h = 0$). The pressure exerted on the trapped air at this point is equal to the atmospheric pressure p

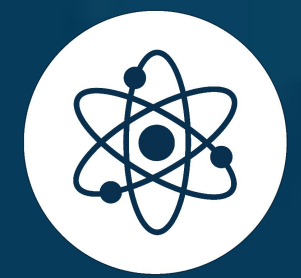


PROCEDURE

2. Raise the right tube by dragging it upward for $h = 50 \text{ mm} = 5 \text{ cm} = 0.05 \text{ m}$. This will cause an increase in the pressure exerted on the trapped air.
3. Notice that the volume occupied by the trapped air has decreased.
4. Click the Record button to record this value of the volume, V_f
5. Raise the right tube again to $h = 100 \text{ mm}$ and record the corresponding value of the volume occupied by the trapped air for each height.
6. Once you finish, an Excel sheet will be downloaded containing the recorded data.

$h \text{ (mm)}$	$V \text{ (10}^{-3}\text{L)}$
0	20.00
20	19.48
50	18.36

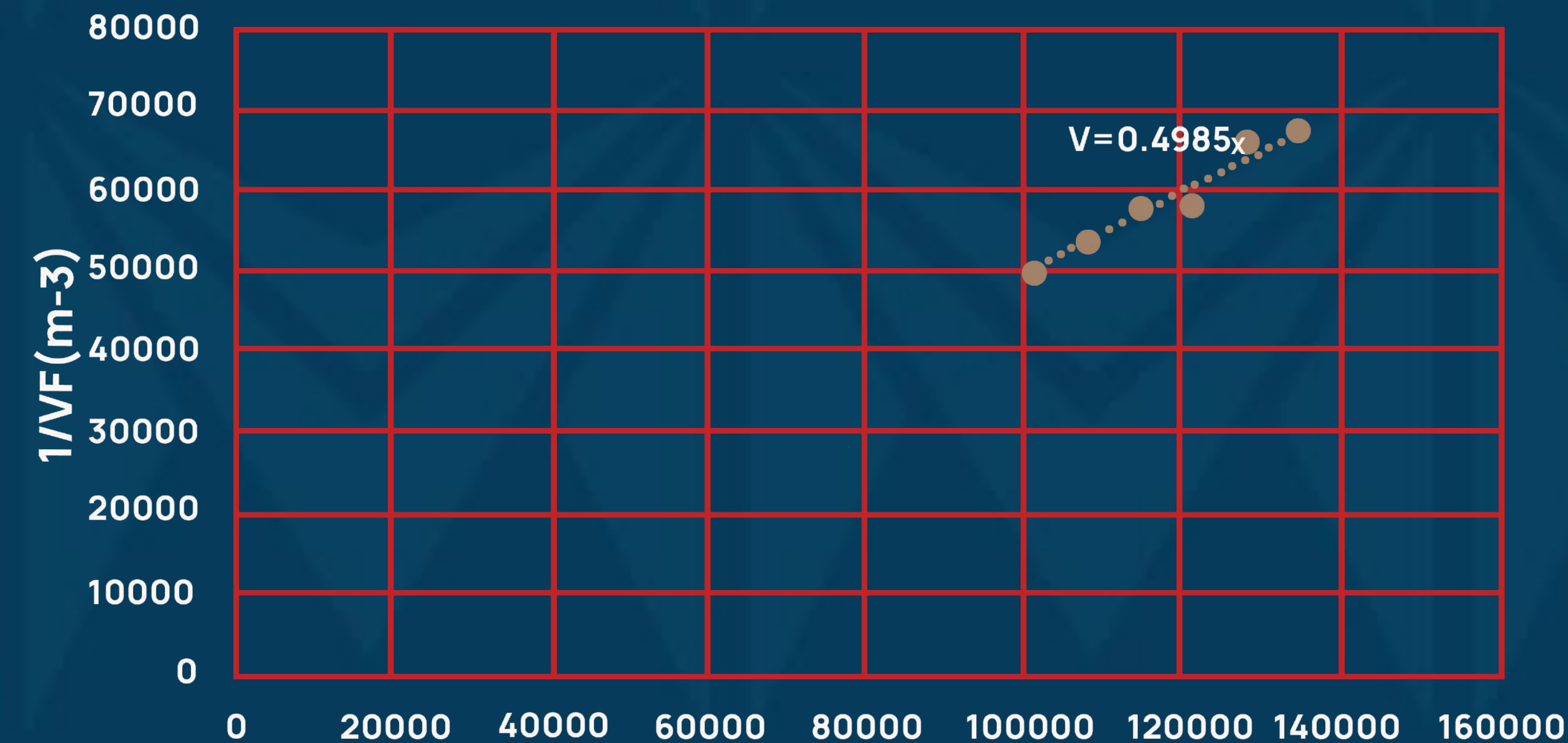




DATA ANALYSIS

1. Make the necessary calculations to convert the values of p_f and V_f to the SI system of units.
2. Plot a graph between $1/V_f$ versus p_f which should be a straight line graph that passes through the origin.
3. Calculate the slope of the graph.
4. Using the equation below, compare your slope from the graph with actual slope from the equation.

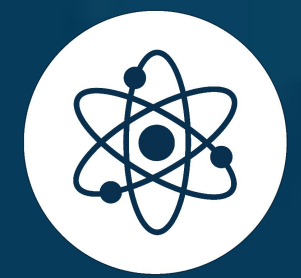
$$\frac{1}{v_f} = \frac{p_f}{2.026}$$





WALK_THROUGH VIDEO





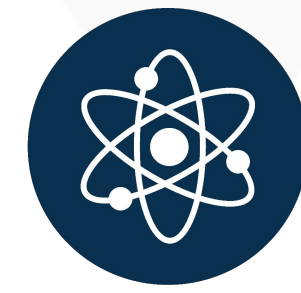
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Sang, D., Hewlett, C., Jones, G., Field, S., & Styles, D. (2020). Cambridge International AS & A Level Physics Practical Workbook. Cambridge University Press.

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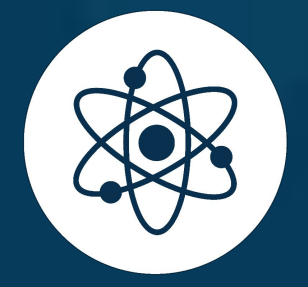
FURTHER READING

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<http://hyperphysics.phy-astr.gsu.edu>

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THANK
YOU